

I-01-CX-frequency-scale-time-kill-test v1: Frequency-scale-time $\Rightarrow$ upper bound on W_+

ZFL-CX

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1 Mission

$$\boxed{T^* < \infty \Rightarrow C_I}$$

$$\boxed{C_E^{weak} \wedge C_{L^3} \wedge C_F^{conc} \wedge C_I \Rightarrow \perp}$$

$$\boxed{\text{Priority: } W_+ \leq f(C_I)}$$

$T^* < \infty$ strong solution blow-up. $W(t) = \int_{\mathbb{R}^3} \det(\nabla u)$? or general vorticity/strain interaction.
 $W_+ = \max(W, 0)$. Known $C_E^{weak} : \limsup \int_0^t W = +\infty \Rightarrow \int_0^{T^*} W_+ = +\infty$, $C_{L^3} : \limsup \|u\|_{L^3} = +\infty$, $C_F^{conc} : \exists t_n \uparrow T^*, x_n, r_n \downarrow 0, \|u(t_n)\|_{L^3(B_{r_n})} \geq \varepsilon_0$.

Goal: find C_I frequency-scale-time such that W_+ controlled.

2 Definition C_I attempt

Let $P_{\leq N}, P_{>N}$ Littlewood-Paley. Define critical frequency envelope:

$$N_c(t) := \inf\{N : \|P_{>N}u(t)\|_{L^3} \leq \varepsilon_0/2\}$$

If $T^* < \infty$, define

$$C_I : \limsup_{t \uparrow T^*} N_c(t) = +\infty \quad \text{and} \quad \int_{T^*-\delta}^{T^*} \|P_{\leq N_c(t)}u\|_{\dot{B}_{2,\infty}^{1/2}}^2 dt < \infty?$$

Or: there exists frequency cascade: $\exists N_n \rightarrow \infty, t_n \uparrow T^*$ with $\|P_{N_n}u(t_n)\|_{L^3} \geq \eta_0$ universal.

Formal proposed:

$$C_I^{freq} : \exists \eta_0 > 0 \text{ univ}, \exists N_n \rightarrow \infty, t_n \uparrow T^*, \|P_{N_n}u(t_n)\|_{L^3} \geq \eta_0$$

Second variant scale-time: $r_n = N_n^{-1}$ links to C_F^{conc} .

3 Claim $W_+ \leq f(C_I)$ attempt

We attempt: $W(t) = \int \omega \cdot S\omega$ (?) decomposition $W = W_{\leq N} + W_{>N} + W_{mix}$.

Goal inequality:

$$W_+(t) \leq C\|u(t)\|_{L^3} N_c(t)^2 + C N_c(t)^{-1} \|\nabla u(t)\|_{L^2}^2?$$

Or via paraproduct: $\det(\nabla u)$ trilinear. Estimate:

$$|W(t)| \lesssim \sum_{N_1 \sim N_2 \sim N_3} N_1^? \|P_{N_1} u\| \|P_{N_2} u\| \|P_{N_3} u\|$$

If C_I forces mass at N_n , then concentration $r_n \sim N_n^{-1}$ matches C_F^{conc} , but does not yet bound W_+ above.

4 Kill Test

I-01 KILLED if any holds:

1. C_I is reformulation of C_{L^3} : C_I^{freq} follows from $\limsup \|u\|_3 = \infty$ without new scale-time info. Then no new bridge. 2. C_I only shows additional concentration: $N_n \rightarrow \infty$ equivalent to $r_n \downarrow 0$ already in C_F^{conc} . No W_+ link. 3. C_I reproduces ε -regularity known: smallness of $\|P_{>N} u\|_{L^3}$ implies regularity, i.e., ESS/Seregin again. 4. Critical norm can diverge while W_+ diverges arbitrarily: no inequality $W_+ \leq f(\|P_{\leq N_c} u\|, \dots)$; f can be $+\infty$ even if C_I holds. 5. C_I requires hypothesis not derived from $T^* < \infty$: e.g., Type I, $\sup(T^* - t)^{1/2} \|u\|_\infty$, a priori bound on profile, extra Besov bound. 6. No inequality connecting frequency/scale-time info to W_+ : i.e., $\int_0^{T^*} W_+$ remains compatible with C_I .

Check 1-6: At present draft, C_I^{freq} is directly implied by C_{L^3} plus frequency envelope; it does not add control of W_+ . The trilinear estimate gives $|W| \lesssim \|u\|_{BMO} \|\nabla u\|_2^2$, but $\|u\|_{BMO}$ not controlled by C_I alone. So $W_+ \leq f(C_I)$ fails.

Hence C_I as defined is not quantitative for W_+ .

5 Conclusion

Thus

$$C_E^{weak} \wedge C_{L^3} \wedge C_F^{conc} \text{ compatible, no } \perp$$

$$C_E^{weak} \wedge C_{L^3} \wedge C_F^{conc} \wedge C_I^{freq} \text{ compatible, no } \perp$$

$$C_I^{freq} \text{ strictly stronger than } C_{L^3} \text{ in frequency, but does not bound } W_+$$

Classification: OPEN as frequency-scale-time is genuinely new family distinct from Lagrangian, but KILLED as bridge to $\perp$ since no $W_+ \leq f(C_I)$ proved.

$\mathcal{C}_{PROVEN}$ remains $\{C_E^{weak}, C_{L^3}, C_F^{conc}\}$. No new $\perp$.

Next: decide if I-01 merits I-02 or move to J-family (genuinely new: e.g., geometric depletion, critical Besov depletion, quantitative CKN L^3 smallness propagation).